# ICS 6B F26 Take Home Exam 6

Due: November 6, 2026 at 11:59PM

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UCI NetID :

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(**alpha-numeric; NOT your student ID**)

- Read the instructions of each question carefully.
- Show your work. Where applicable, partial credit will be given on the work shown.
- Please keep in mind the academic honesty guidelines. This take-home exam is to be completed individually, with no outside help. You may use any resources from our class (ZyBooks and resources from Canvas), but you may not use any other online resources.
- You may choose to print the exam or use a digital editor for completing the exam. It is required that you use this PDF to complete your work. If you have no access to a printer or digital tools to fulfill the exam, feel free to reach out to the staffs regarding your concerns.
- If you have any questions, please post a private Ed or attend available Office Hours. Note that we are not allowed to provide specific help to answering the exam questions.
- There are **6** problems on this test.
- There are a total **14** of points.
- Good luck!

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# Problem 1 (2 points)

Rewrite the logical expression "$(a \wedge b) \leftrightarrow (c \vee d)$" as a Boolean expression of the Boolean variables $a, b, c$ and $d$ (the allowed Boolean operations include multiplication, addition and complement). You do not need to simplify the expression any further than what's necessary to use only the basic Boolean operators.

We know

$X \leftrightarrow Y = (X \wedge Y) \vee (\neg X \wedge \neg Y)$.

Thus

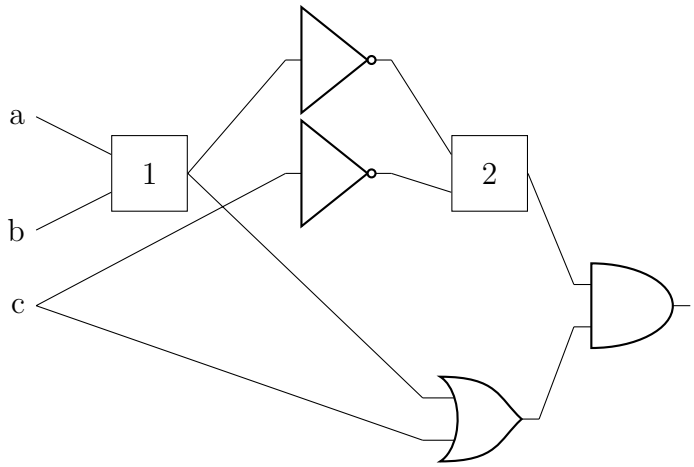
$(a \wedge b) \leftrightarrow (c \vee d) = ab(c + d) + (\neg a + \neg b)(\neg c \wedge \neg d)$.

Equivilantly

$abc + abd + \neg a\,\neg c\,\neg d + \neg b\,\neg c\,\neg d$.

# Problem 2 (1 points)

Choose the right gates for the boxes 1 and 2 to make the following circuit equivalent to the Boolean expression representation of the logical expression '$(a \vee b) \leftrightarrow c$' (assume the wires at the intersections go straight instead of making turns, so the top line from input $c$ goes to the inverter and not the OR gate.). Choose 'None of the others' if none of the options make the corresponding circuit to the Boolean expression above.




◯ Box 1: OR, Box 2: AND ◯ Box 1: AND, Box 2: OR
◯ Box 1: AND, Box 2: AND ◯ Box 1: OR, Box 2: OR
✓ ◯ None of the others

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# Problem 3 (3 points)

You are given the following table for a Boolean function $f(x, y, z)$:

| $a$ | $b$ | $c$ | $f(a,b,c)$ | $\overline{f(a,b,c)}$ |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 | 1 |

Fill in the table and give the DNF of $\overline{f(a,b,c)}$ i.e. represent the complement of $f$ as a sum of minterms.

$\overline{f(a,b,c)} = \bar{a}b\bar{c} + \bar{a}bc + a\bar{b}c + ab\bar{c} + abc$

Give the CNF of $f$.

$f(a,b,c) = (a + \bar{b} + c)(a + \bar{b} + \bar{c})(\bar{a} + b + \bar{c})(\bar{a} + \bar{b} + c)(\bar{a} + \bar{b} + \bar{c})$

# Problem 4 (4 points)

Identify whether the following functions are in DNF, CNF, both or neither.

1. $\overline{w}xy\overline{z}$
   ◯ DNF ◯ CNF ✓ Both ◯ Neither
2. $xyz + w\overline{xyz} + wy\overline{z}$
   ◯ DNF ◯ CNF ◯ Both ✓ Neither
3. $(wxy + \overline{z})x\overline{y}$
   ◯ DNF ◯ CNF ◯ Both ✓ Neither
4. $(w + \overline{x} + y)(w + \overline{y} + \overline{z})z$
   ◯ DNF ✓ CNF ◯ Both ◯ Neither

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# Problem 5 (2 points)

Choose whether the following set of expressions over some symbols $\{\div, \blacklozenge\}$ shows they are functionally complete(without any additional manipulation). Briefly explain your solutions.

1. $(x \div x) \blacklozenge x = \overline{x}$ and $x \div (y \div (x \blacklozenge y)) = xy$

   ◯ Functionally Complete ◯ Not Functionally Complete

   Explain:

   The first identity, $(x \leftrightarrow x) \blacklozenge x = \overline{x}$ provides an expression for the logical NOT operation.

   The second identity, $x \leftrightarrow (y \leftrightarrow (x \blacklozenge y)) = xy$ provides an expression for the logical AND operation.

   *Since the set* $\{\text{NOT}, \text{AND}\}$ *is functionally complete, the set of symbols* $\{\leftrightarrow, \blacklozenge\}$ *is also functionally complete.*

2. $(x \blacklozenge y) \blacklozenge (y \div \overline{z}) \div (\overline{x} \blacklozenge z) = x \downarrow y$

   ◯ Functionally Complete ◯ Not Functionally Complete

   Explain:

   Because the NOR operation $\downarrow$ is a functionally complete set by itself, the ability to express it (as shown in the equation $(x \blacklozenge y) \blacklozenge (y \leftrightarrow \overline{z}) \leftrightarrow (\overline{x} \blacklozenge z) = x \downarrow y$) demonstrates that the set of symbols $\{\leftrightarrow, \blacklozenge\}$ is functionally complete.

# Problem 6 (2 points)

A baseball game has three players in the outfield. $L_i$ represents the left outfielder making a catch during the $i$-th inning. Similarly, $C_i$ represents the center outfielder and $R_i$ represents the right outfielder. Consider an extended game with 18 innings. Define set $A = \{1, 2, \ldots, 18\}$. Write a quantified Boolean expression(i.e. statements of the form $\forall i \in A$ or $\exists j \in A$) over $A$ that evaluates to 1 if and only if the left and center outfielders make all their catches during the same inning(s) but the right outfielder makes all their catches during a different inning(s).

$(\forall i \in A\ (L_i \leftrightarrow C_i)) \wedge (\forall i \in A\ (R_i \rightarrow (\neg L_i \wedge \neg C_i)))$ is a quantified Boolean expression